CODEBASE ENGINEERING BLUEPRINT · SENIOR DEVELOPER SPECIFICATION

# AptRun Technical Feature Guide & Codebase Architecture

The definitive codebase manual for Senior Developers, Architects, and Tech Leads. Details database schemas, plan limits & entitlement logic, background workers, judge execution isolation, AI integration pipelines, and extension points to guide feature modifications and system enhancements.

**STATUTORY CORPORATE REGISTRY & INTELLECTUAL PROPERTY ATTESTATION** MCA STATUTORY RECORD LEGALLY VALIDATED

| PLATFORM PRODUCT | PROPRIETARY BRAND | PARENT CORPORATE BODY | CORPORATE IDENTITY NO. (CIN) |
|---|---|---|---|
| **AptRun™ Platform (aptrun.com)** | **Mellow Vault (mellowvault.com)** | **PRAYUKTI DEVELOPMENT PRIVATE LIMITED** | U62099UP2024PTC207212 |
| **STATUTORY AUTHORITY** | **REGISTERED OPERATIONS BASE** | **ENTERPRISE TECH DESK** | **SECURITY & COMPLIANCE** |
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## 1. Platform Architectural Topology

**Next.js 14**
FRONTEND LAYER
App router, Monaco Editor, Tailwind CSS, KaTeX, Framer Motion

**Laravel 11**
CONTROL PLANE
PHP 8.2+, Sanctum RBAC, Eloquent ORM, MySQL/Postgres

**Piston Judge**
EXECUTION CLUSTER
19 language runtimes, 256MB cgroup caps, sub-second latency

**Caddy + Docker**
PROJECT LAB HOSTS
Hardened OpenVSCode Server containers with reverse wildcard TLS

**Core Structural Invariant: Untrusted Code Isolation**
Student code submissions and container workspaces **never run on the PHP web application server**. The Laravel application functions strictly as a metered, secure control plane. Piston executes code in isolated execution cgroups, while Project Lab workspaces run inside unprivileged Linux containers ( `--cap-drop ALL` , `no-new-privileges` ) on dedicated worker nodes.

## 2. Repository Directory Structure & Subsystems

| DIRECTORY PATH | TECHNOLOGY STACK | ARCHITECTURAL FUNCTION & CODE RESPONSIBILITIES |
|---|---|---|
| `backend/app/Models/` | Eloquent ORM (PHP) | Contains models: `User` , `Plan` , `Subscription` , `Problem` , `Contest` , `CoreQuestion` , `Workspace` . Defines relationships, scopes, and casts. |
| `backend/app/Services/` | Business Logic Layer | Domain services: `ReadinessScoreService` , `PurchaseQuoteService` , `CheckoutService` , `WorkspaceManager` , `SpeechAlignment` , `VocabularySrsService` . |
| `backend/routes/api.php` | Routing & Middleware | Central routing topology partitioned by role middleware ( `role:user` , `role:admin_tpo` , `role:superadmin` , `account.active` , `not.impersonating` ). |
| `frontend/src/app/` | Next.js App Router | Contains all persona routes: `/dashboard` (Students), `/admin` (TPO/Ops), `/coordinator` (Faculty), `/superadmin` (Governance), `/arena` (Hackathons). |
| `infra/judge/` & `workspaces/` | Docker, Caddy, Bash | Supervisor scripts, Piston runtime patches, Caddy wildcard gateway configuration, and OpenVSCode container Dockerfiles. |

## 1. Role Constants & Tenant Boundaries

Authorization is defined via role constants on the `User` model and enforced via route middleware. Cross-tenant access attempts return `404 Not Found` (rather than 403) to prevent enumerating the existence of other institutions or companies.

| ROLE CONSTANT | STRING TOKEN | SCOPE OF AUTHORITY & ACCESSIBLE CONTROLLERS | TENANT BOUNDARY RULE |
|---|---|---|---|
| `ROLE_USER` | `user` | Coding console, core prep, learning lab, project lab, AI interview sessions, vault, drive applications. | Sees personal progress and college-mapped content only. |
| `ROLE_ADMIN_TPO` | `admin_tpo` | Batch onboarding, 29-col cohort directory, mock assessments, anti-cheat proctoring, drive mapping, reports. | Scoped strictly to `users.college_id`. |
| `ROLE_SECTION_COORDINATOR` | `section_coordinator` | Sub-cohort attention queue, daily intervention logging, section mock proctoring, disciplinary blocks. | Scoped strictly by `users.coordinator_scope` (Section + Branch). |
| `ROLE_ADMIN_COMPANY` | `admin_company` | College directory proposals, job openings, talent pool search, custom coding tests, AI interview reviews. | Scoped strictly to `users.company_id`. Other drives return 404. |
| `ROLE_ADMIN_INTERNAL` | `admin_internal` | Platform operations: Problem bank curation, 70 soft skill sets, Core Prep import, customer success. | Governed by granular permission flags in `users.permissions`. |
| `ROLE_SUPERADMIN` | `superadmin` | Platform pricing, subscription overrides, payment gateway audit, user impersonation, system flags. | Unrestricted platform access; immutable audit logging. |

## 2. Internal Staff Granular Permissions ( `PERMISSIONS_INTERNAL` )

`PERM_PROBLEM_BANK` : Author/edit DSA challenges, test cases, and editorial approaches.

`PERM_CORE_PREP` : Import ECE/EEE/ME/Civil banks, author tests, toggle go-live switches.

`PERM_SOFT_SKILLS` : Create/edit 25-question test papers across Sets 1 to 70.

`PERM_PROJECT_LAB` : Author SOW blueprints, configure Docker workspace images and rubrics.

`PERM_HACKATHONS` : Create and judge platform-wide or college hackathons in `/arena`.

`PERM_VAULT` : Inspect student proof-of-work and manage Mellow Vault enterprise sync.

## 3. Critical Middleware Stack in `routes/api.php`

| MIDDLEWARE NAME | CLASS FILE | SECURITY ENFORCEMENT LOGIC |
|---|---|---|
| `account.active` | `EnsureAccountIsActive` | Checks `users.is_active`. If blocked by a coordinator, TPO, or admin, terminates request with 403 and the exact blocking reason. |
| `not.impersonating` | `RejectWritesWhileImpersonating` | If token has `IMPERSONATION_ABILITY`, intercepts all mutating HTTP verbs (POST, PUT, DELETE, PATCH) and aborts with 403. |
| `auth.vault` | `AuthenticateVaultIntegration` | Validates shared secret bearer token on `/api/vault/v1/*` routes for Mellow Vault synchronization. |

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## 1. Database Schema: `plans` Table

The `plans` table defines commercial models, audiences, access flags, and numeric rate-limiting columns enforced throughout the API:

| COLUMN NAME | DATA TYPE | DEFAULT VALUE | FUNCTIONAL ROLE & CODE ENFORCEMENT |
|---|---|---|---|
| `code` | `VARCHAR(50)` | — (Unique) | Unique string identifier: `coder`, `expert`, `grandmaster`, `standard`, `pro-campus`. |
| `audience` | `ENUM` | `individual` | `individual` (students), `institution` (colleges), `company` (recruiters). |
| `kind` | `ENUM` | `base` | `base` (primary subscription; 1 per user) vs `addon` (stackable module). |
| `list_price` | `INT NULL` | `0` | Price in INR before discount. `0` = free plan; `NULL` = contact-only. |
| `discount_percent` | `INT` | `0` | Discount percentage (max 90%). `salePrice() = list_price - discount`. |
| `annual_price` | `INT NULL` | `NULL` | Annual contract price for institutional tiers (e.g. ₹1,49,000 for Standard). |
| `duration_days` | `INT NULL` | `NULL` | Subscription period: 365 (Expert, College), 1095 (Grandmaster), `NULL` (Free Coder). |
| `contact_only` | `BOOLEAN` | `FALSE` | If true, self-serve purchase is blocked; sold offline by team (Grandmaster). |
| `max_practice_problems_per_day` | `INT NULL` | `NULL` | Daily coding solve quota. `1` for Coder; `NULL` = unlimited. |
| `max_mock_interviews_per_day` | `INT NULL` | `NULL` | Daily AI interview quota. `10` for Expert; `NULL` = unlimited. |
| `feature_trial_uses` | `INT NULL` | `NULL` | Lifetime free tries for AI features on free plan (default `1` for Coder). |
| `project_lab_access` | `BOOLEAN` | `FALSE` | Unlocks OpenVSCode container workspaces (Expert, Grandmaster, College). |
| `max_workspace_hours_per_month` | `INT NULL` | `NULL` | Cap on active container runtime hours per calendar month (default: 80 hrs). |
| `max_active_projects` | `INT NULL` | `NULL` | Max concurrent active projects (default: 3 projects). |
| `vault_access` | `BOOLEAN` | `FALSE` | Allows profile listing in Mellow Vault when ARS ≥ 60. |
| `hackathon_access` | `BOOLEAN` | `TRUE` | Allows entering and submitting projects in Hack Arena (`/arena`). |
| `drive_access` | `BOOLEAN` | `FALSE` | Allows viewing and registering for campus placement drives. |
| `max_students` | `INT NULL` | `NULL` | Seat limit for institution plans (Standard: 500, Pro Campus: 2000, Enterprise: NULL). |
| `max_core_practice_per_day` | `INT NULL` | `NULL` | Daily MCQ practice quota for non-CS engineering branches (Coder: 10, Expert: NULL). |
| `core_tests_access` | `BOOLEAN` | `FALSE` | Allows taking timed, proctored department engineering mock exams. |

## 2. Entitlement Resolution Flow: `User::effectiveEntitlements()`

**Precedence Logic: Institutional vs Individual Plans**

When resolving a student's active limits:
1. If student belongs to a subscribed College (`effectiveSubscription()` resolves to college), the **College Plan governs** (unlimited practice, interviews, drives, core prep).
2. If college has no active subscription, the student's **Personal Paid Plan (Expert/Grandmaster)** applies.
3. If no paid plan is active or if end date has passed, the system falls back to the default **Free Coder** plan.

## 1. Individual Plan Comparison Matrix

The individual catalogue governs self-registered students and students purchasing personal upgrades outside their college:

| FEATURE DIMENSION | CODER (FREE PLAN) | EXPERT (ANNUAL PAID) | GRANDMASTER (MELLOW ACADEMY) |
|---|---|---|---|
| **Commercial Term** | Free on signup; never expires. | 1 Year (365 days); self-serve checkout. | 3 Years (1,095 days); sold offline by team. |
| **List & Sale Price (INR)** | ₹0 / lifetime | ₹4,999 list / ₹2,999 sale (configurable) | Arranged with admissions team (no online price) |
| **Coding Practice Quota** | **1 problem / day** (throttle: `practice.daily`) | **Unlimited** (19 language runtimes) | **Unlimited** (19 language runtimes) |
| **AI Mock Interviews** | **1 lifetime free try** (trial flag) | **10 interviews / day** | **10 interviews / day** |
| **Learning Centre AI Features** | 1 lifetime try of Speaking, AI passage, AI quiz | **Unlimited** daily practice | **Unlimited** daily practice |
| **Project Lab (Browser VS Code)** | LOCKED (upgrade prompt) | **80 hrs/mo** · 3 active projects | **80 hrs/mo** · 3 active projects |
| **The Mellow Vault Access** | LOCKED (unlocks at ARS 60) | **Yes** (unlocked when ARS ≥ 60) | **Yes** (unlocked when ARS ≥ 60) |
| **Hack Arena Access** | NO | **Yes** (participate & win prizes) | **Yes** (participate & win prizes) |
| **Campus Placement Drives** | NO | **Yes** (apply to corporate drives) | **Yes** (apply to corporate drives) |
| **Core Engineering Prep** | 10 MCQs/day · No timed tests | **Unlimited** · Timed proctored mock tests | **Unlimited** · Timed proctored mock tests |
| **Development Curriculum** | None | None | **MERN, Laravel, Full Stack, Mobile** |
| **Placement Guarantee** | None | None | **Assured Placement Commitment** |

## 2. Code Safeguards for Individual Plans

**The Grandmaster Contact-Only Lock**

`plans.contact_only = true` : `PurchaseQuoteService` checks this before reading any price and rejects self-serve checkout with `422 Unprocessable Entity`. Grandmaster cannot be purchased online even if an attacker manipulates the payload. It must be granted manually by Superadmin.

**The Retired Project Lab Add-On ( `project-lab` )**

Project Lab was originally sold as an add-on (₹499/mo). It is now bundled into Expert and Grandmaster. The row remains in `plans` with `is_active = false` to honor legacy active terms until expiration, while preventing new add-on sales.

## 3. Rate-Limiting Implementation Details

```
// Example: Checking daily coding limit in ProblemController.php
$entitlements = $user->effectiveEntitlements();
if ($entitlements['max practice problems per day'] !== null) {
    $todayCount = Submission::where('user_id', $user->id)
        ->where('created at', '>=', now()->startOfDay())
        ->where('status', 'accepted')
        ->count();
    if ($todayCount >= $entitlements['max practice problems per day']) {
        return response()->json(['error' => 'daily_limit_reached', 'upgrade_url' => '/dashboard/billing'], 429);
    }
}
```

## 1. Institutional Plan Comparison Matrix

Institutional plans ( `audience = 'institution'` ) are sold by Mellow Ops to engineering colleges and universities. All covered students receive unlimited practice without personal subscriptions:

| FEATURE DIMENSION | STANDARD COLLEGE PLAN | PRO CAMPUS PLAN | ACADEMIC ENTERPRISE PLAN |
|---|---|---|---|
| **Target Institution Size** | Single department or mid-sized college | Large engineering college (multiple branches) | Multi-campus university or academic group |
| **Student Seat Capacity** | **500 Students** ( `max_students = 500` ) | **2,000 Students** ( `max_students = 2000` ) | **Unlimited Seats** ( `max_students = NULL` ) |
| **Annual Contract Price** | ₹1,49,000 / year ( `annual_price` ) | ₹4,49,000 / year ( `annual_price` ) | Custom Enterprise SLA (tailored pricing) |
| **Student Practice & Mocks** | **Unlimited** practice & mock tests | **Unlimited** practice & mock tests | **Unlimited** practice & mock tests |
| **Campus Drives Portal** | **Yes** · Map catalog drives & host private | **Yes** · Map catalog drives & host private | **Yes** · Multi-campus pooled drives |
| **Proctored Mock Contests** | Webcam, fullscreen lock, 3-strike rule | Webcam, fullscreen lock, 3-strike rule | **AST-based anti-cheat & proctoring** |
| **Section Coordinators** | Up to 10 coordinators | Up to 40 coordinators | **Unlimited Coordinators & HODs** |
| **Core Prep (ECE, EEE, ME, Civil)** | Basic topic practice | Full Core Prep + Timed Tests | **Full Core Prep + Custom Subject Banks** |
| **The Vault & Hack Arena** | Included for all covered students | Included for all covered students | Included for all covered students |
| **Placement Intelligence** | 7 standard Excel & PDF reports | 7 reports + branch-wise comparative heatmaps | **7 reports + Custom ERP/LMS integrations** |
| **Support & Account Management** | Standard Email & Ticket Support | Priority Support + Dedicated Onboarding | **Dedicated Success Manager & Custom SLA** |

## 2. How Seat Limits ( `max_students` ) Are Enforced in Code

The student onboarding controller validates seat caps during bulk CSV imports and individual self-registrations:

```
// In TpoStudentController.php & BulkImportService.php
$college = $tpo->college;
$plan = $college->activeSubscription?->plan;

if ($plan && $plan->max_students !== null) {
    $currentActiveCount = User::where('college_id', $college->id)
        ->where('role', User::ROLE_USER)
        ->count();

    if (($currentActiveCount + count($importRows)) > $plan->max_students) {
        return response()->json([
            'error' => 'seat_limit_exceeded',
            'message' => "Your plan ({$plan->name}) caps enrolled students at {$plan->max_students}. Current: {$currentActiveCount}."
        ], 422);
    }
}
```

## 3. College Add-on Modules & Future Entitlements

**Extending Project Lab to College Plans**

By default, `plans.project_lab_access = false` on institutional rows. To grant an entire college Project Lab access (e.g. for a dedicated Full-Stack Capstone semester), the Superadmin simply checks `project_lab_access = true` on that college's plan row in Superadmin Master Control. All covered students receive container workspaces instantly.

## 1. Recruiter Subscription Models ( `audience = 'company'` )

Enterprise hiring partners subscribe to dedicated corporate tiers to recruit pre-assessed university engineering talent:

| RECRUITMENT CAPABILITY | HIRING STARTER (₹99,000 / YEAR) | HIRING GROWTH (₹2,99,000 / YEAR) |
|---|---|---|
| **Active Job Openings** | Up to **3 Concurrent Openings** | **Unlimited Active Openings** |
| **Campus Drive Proposals** | Up to 10 Partner Colleges / year | **Unlimited Partner Colleges** |
| **Pre-Assessed Talent Pool Search** | Basic search (Branch, College Tier, ARS range) | **Deep search** (Code quality, Project Lab badges, Git repos) |
| **Direct Candidate Outreach** | 50 Direct Interview Invites / month | **300 Direct Interview Invites / month** |
| **Custom Coding Assessments** | Standard shared problem bank | **Proprietary problem statements + private test suites** |
| **Asynchronous AI Interview Review** | Transcripts only | **Full voice audio playback + synchronized transcript + rubric overrides** |
| **Anti-Cheat & Plagiarism Analysis** | Webcam snapshots & tab blur counts | **Full keystroke replay timeline + AST cross-candidate similarity** |
| **Tamper-Proof Vault Verification** | Summary report | **Full cryptographic verification & source code inspection** |
| **ATS Integration Export** | Manual CSV export | **Automated webhook export (Greenhouse, Workday, Lever)** |

## 2. Company Tenant Isolation Architecture

**Strict Multi-Tenant Segregation in `CompanyController.php`**

Every corporate recruiter belongs to a `Company` tenant model. The base query scope automatically binds queries to `company_id`:

```
// Enforced in CompanyBaseController.php
public function scopeOwn(Builder $query) {
    return $query->where('company_id', auth()->user()->company_id);
}
```

If a recruiter attempts to access an assessment or drive ID belonging to a competing firm, the controller aborts with `404 Not Found`.

## 3. Proposed Extensions for Corporate Plans

**Extension Point 1: Per-Seat Recruiter Licenses**

Currently, corporate accounts have one primary login. A recommended enhancement is adding a `company_users` table with roles: *Hiring Admin*, *Technical Interviewer*, and *Sourcer*.

**Extension Point 2: Custom Branded Assessment Portals**

Adding a `whitelabel_domain` and `brand_theme` column on `companies` to allow enterprise recruiters to serve tests under `careers.company.com`.

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## 1. Core Entity Relationship Topology

The database schema is organized into modular bounded contexts: Identity & Auth, Assessment Engine, Learning Centre, and Container Workspaces.

| TABLE NAME | PRIMARY KEY & FOREIGN KEYS | CORE SCHEMA COLUMNS & INDEXES |
|---|---|---|
| `users` | `id (PK)`, `college_id (FK)`, `company_id (FK)`, `discipline_id (FK)` | `role`, `name`, `email`, `password`, `roll_number`, `branch`, `section`, `cgpa`, `backlogs`, `coordinator_scope (JSON)`, `track_focus`, `is_active`, `permissions (JSON)`. |
| `colleges` | `id (PK)` | `name`, `slug`, `code`, `tier`, `city`, `state`, `logo_url`, `is_active`, `max_students_override`. |
| `plans` | `id (PK)`, `pricing_updated_by (FK)` | `code`, `name`, `audience`, `kind`, `list_price`, `discount_percent`, `annual_price`, `duration_days`, `max_practice_problems_per_day`, `max_mock_interviews_per_day`, `project_lab_access`. |
| `subscriptions` | `id (PK)`, `user_id (FK)`, `college_id (FK)`, `plan_id (FK)` | `status (active/cancelled/expired)`, `starts_at`, `ends_at`, `cancelled_at`, `price_paid`, `meta (JSON)` (stores payment ref, source: admin_grant vs gateway). |
| `problems` | `id (PK)` | `title`, `slug`, `difficulty (easy/medium/hard)`, `description`, `input_format`, `output_format`, `constraints`, `time_limit_ms`, `memory_limit_mb`, `tags (JSON)`. |
| `submissions` | `id (PK)`, `user_id (FK)`, `problem_id (FK)`, `contest_id (FK)` | `language`, `code`, `status (AC/WA/TLE/MLE/CE)`, `execution_time_ms`, `memory_consumed_kb`, `test_cases_passed`, `total_test_cases`. |
| `contests` | `id (PK)`, `college_id (FK)`, `company_id (FK)` | `title`, `slug`, `type (daily/rated/mock/hiring)`, `starts_at`, `ends_at`, `duration_minutes`, `is_proctored`, `audience (JSON)`. |
| `workspaces` | `id (PK)`, `user_id (FK)`, `project_id (FK)` | `container_id`, `host_server`, `port`, `status (running/stopped/reaped)`, `last_heartbeat_at`, `total_seconds_used`, `volume_name`. |
| `core_questions` | `id (PK)`, `discipline_id (FK)`, `branch_id (FK)` | `question_text`, `options (JSON)`, `correct_option`, `explanation`, `topic_slug`, `difficulty`, `is_published`. |
| `audit_logs` | `id (PK)`, `actor_id (FK)` | `action`, `target_type`, `target_id`, `diff (JSON)`, `reason`, `ip_address`, `user_agent`, `created_at`. |

## 2. Critical Database Indexing Strategy

- `submissions(user_id, status, created_at)`: Optimized for ARS consistency and practice breadth aggregation.
- `users(college_id, section, branch)`: Accelerates TPO cohort queries and Section Coordinator scoping.
- `workspaces(user_id, status)`: Used by container reaper daemon to terminate idle containers.

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## 1. Judge Pipeline & Container Isolation Model

The AptRun Code Judge executes student solutions across 19 runtimes using a cluster of Piston execution engines. Execution is stateless, sandboxed, and enforced via Linux control groups (cgroups).

**256 MB**
**MEMORY CAP**
Strict cgroup memory quota prevents OOM host crashes

**1.5 Cores**
**CPU QUOTA**
Throttles infinite CPU spinloops and forkbombs

**2.0 Seconds**
**WALL-CLOCK TIMEOUT**
Automated SIGKILL terminates execution past deadline

**No Network**
**AIR-GAPPED SANDBOX**
`--network none` blocks outbound socket requests

## 2. Supported Programming Language Matrix

| LANGUAGE | PISTON RUNTIME | COMPILER / VM | STANDARD EXECUTION FLAGS & COMPILATION PARAMETERS |
|---|---|---|---|
| **C** | `c (10.2.0)` | GCC 10.2 | `gcc -O3 -std=c11 solution.c -lm` |
| **C++** | `cpp (10.2.0)` | G++ 10.2 | `g++ -O3 -std=c++17 solution.cpp -lm` |
| **Java** | `java (15.0.2)` | OpenJDK 15 | `javac Solution.java && java -Xmx256M -Xss64M Solution` |
| **Python 3** | `python (3.10.0)` | CPython 3.10 | `python3 -u solution.py` (unbuffered stdout) |
| **Go** | `go (1.16.2)` | Go Compiler | `go build -o solution solution.go && ./solution` |
| **Rust** | `rust (1.50.0)` | rustc | `rustc -O solution.rs && ./solution` |
| **TypeScript** | `typescript (4.2.3)` | Node.js 16 + tsc | `esbuild solution.ts --outfile=out.js && node out.js` |
| **C#** | `csharp.net (5.0)` | .NET Core 5.0 | `dotnet run --configuration Release` |

## 3. How to Add a New Programming Language (Senior Dev SOP)

**3-Step Extension Procedure**

To add a new language (e.g. **Elixir** or **Zig**):
1. **Install Runtime:** Run `piston-cli install zig` on Piston runner host (`infra/judge/supervisor`).
2. **Register Alias in `JudgeService.php`:** Add `'zig' => ['runtime' => 'zig', 'version' => '0.11.0']` in `SUPPORTED_LANGUAGES`.
3. **Frontend Monaco Binding:** In `frontend/src/components/editor/MonacoEditor.tsx`, add the language mode and default boilerplate starter code.

## 1. Container Workspace Topology

Project Lab provides full-stack containerized workspaces. The Laravel API orchestrates Docker containers on dedicated hosts via SSH, while Caddy handles reverse proxy routing with wildcard TLS certificates.

**Request Routing Architecture**

**Student Browser:**
- Iframe 1 (VS Code IDE): `https://ws-<id>.ws.aptrun.com` → Caddy Gateway → Docker container port `13370` (OpenVSCode Server).
- Iframe 2 (Live App Preview): `https://p<port>--ws-<id>.ws.aptrun.com` → Caddy Gateway → Container user port (e.g. `3000` or `8000`).

## 2. Workspace Lifecycle State Machine

| LIFECYCLE STATE | TRIGGERING EVENT / API | HOST SYSTEM ACTION & RESOURCE STATE |
|---|---|---|
| **CREATING** | `POST /api/workspaces` | Allocates port from pool; initializes Docker volume `wsvol-<id>`; unzips project starter repo. |
| **RUNNING** | Container successfully started | Docker container bound to host port; Caddy dynamic wildcard routing active; heartbeat daemon active. |
| **STOPPED** | User clicks "Stop" or browser closed | `docker stop ws-<id>` executed; port returned to pool; volume remains persisted on disk. |
| **REAPED** | Heartbeat idle > 15 minutes | Scheduled cron (`WorkspaceReaper`) stops container to preserve monthly student hours. |
| **SNAPSHOT_SUBMIT** | User clicks "Submit for Review" | Executes `tar -czf snapshot.tar.gz /home/workspace` inside container; uploads to S3 for review. |

## 3. The Container Dockerfile Blueprint ( `infra/workspaces/Dockerfile` )

```
FROM gitpod/openvscode-server:latest
USER root
# Install Node.js LTS, Python, GCC, Git, and Database CLI tools
RUN apt-get update && apt-get install -y git curl python3 python3-pip sqlite3     && curl -fsSL https://deb.nodesource.com/setup_20.x | bash -
USER openvscode-server
ENV PORT=13370
EXPOSE 13370 3000 8000 8080
ENTRYPOINT ["/bin/sh", "/entrypoint.sh"]
```

## 4. SOW Watermarking Service ( `SowPdfService.php` )

When a student downloads the Statement of Work (SOW PDF), `SowPdfService` generates a dynamic PDF via `dompdf`. The student's name, email, roll number, and cryptographic verification hash are stamped diagonally across each page as an indelible watermark to prevent copyright infringement.

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## 1. Mathematical Formulation & Weights

The Placement Readiness Score (ARS v4) is an objective, deterministic 0–100 rating. It evaluates demonstrated engineering capability over a rolling 180-day recent-form window, scaled by an attempt depth factor to eliminate lucky single attempts.

**The Composite ARS Mathematical Formula**

$$ ext{ARS} = \sum_{k \in ext{Components}} W_k \cdot ext{Score}_k \cdot ext{DepthScaling}(N_k)$$ Where:

- $W_k$: Weight assigned to component $k$ ($\sum W_k = 100$).
- $ ext{Score}_k$: Normalized average score ($0.0 ext{ to } 1.0$) across up to the newest 5 completed attempts within 180 days.
- $ ext{DepthScaling}(N_k)$: Scaling factor $= \min\left(1.0, rac{N_k}{3} ight)$ (Soft Skills uses $rac{N_k}{2}$). A single attempt receives only partial credit.

## 2. Multi-Profile Weighting Matrix ( `config/readiness.php` )

| COMPONENT ($K$) | SOFTWARE ($W_K$) | BALANCED ($W_K$) | CORE ($W_K$) | UNDERLYING DATA TELEMETRY |
|---|---|---|---|---|
| **AI Mock Interviews** | 20 | 20 | 20 | Average of last 5 scored AI interviews. |
| **Core Technical Subject** | — | 20 | 30 | $0.60 imes ext{Mock Tests} + 0.40 imes ext{Topic Mastery}$. |
| **Coding Practice Depth** | 15 | 10 | — | Breadth across Easy (20), Med (40), Hard (40) + topics. |
| **Learning Centre Lab** | 15 | 10 | 15 | Speaking (40%) + Listening (30%) + Vocab SRS (30%). |
| **Contest Elo Rating** | 15 | 5 | — | Share of points scored over last 5 rated contests. |
| **Practice Consistency** | 15 | 15 | 15 | Active practice days in last 28 days (12 days = max score). |
| **Soft Skills Mastery** | 10 | 10 | 10 | Average score across last 5 completed standardized sets. |
| **Events & Projects** | 10 | 10 | 10 | Project Lab submissions, hackathons, and mock exams. |

## 3. The 30-Day Go-Live Grace Period Algorithm

```
// In ReadinessScoreService.php
if ($profile === 'core' && $discipline->isWithinGracePeriod()) {
    $softwareScore = $this->calculateScore($user, 'software');
    $coreScore = $this->calculateScore($user, 'core');
    // Student automatically retains the better score during transition
    return max($softwareScore, $coreScore);
}
```

## 4. Readiness Tiers & Threshold Triggers

**≥ 75 ARS**
PLACEMENT READY
Candidate shortlisted for high CTC campus drives

**45 – 74 ARS**
IN PROGRESS
Eligible for standard campus recruitment

**< 60 ARS**
AT-RISK THRESHOLD
Triggers alert in Section Coordinator queue & TPO reports

## 1. Audio-First Speaking Practice: Technical Pipeline

Speaking Practice coaches verbal fluency without trusting client captions. The audio recording is the sole input, evaluated server-side using Gemini audio models and deterministic speech alignment.

> **Critical Security & Privacy Rule: Zero Audio Retention**
> Student audio buffers are processed entirely in memory. Once the Gemini transcription and `SpeechAlignment` algorithms complete, the audio stream is discarded. **No student voice recordings are saved to disk or persistent storage**.

## 2. Word Alignment Algorithm ( `SpeechAlignment.php` )

| EVALUATION METRIC | COMPUTATION LAYER | MATHEMATICAL FORMULA & SCORING MECHANICS |
|---|---|---|
| **Accuracy (35%)** | Deterministic PHP | Dynamic time-warping compares heard transcript against target passage. Missing or inserted words penalize score. |
| **Clarity (35%)** | Gemini Audio Model | Acoustic model scores phonetic articulation, vowel pronunciation, and ending consonants ($0-100$). |
| **Fluency & Pace (30%)** | Deterministic PHP | Calculates Words Per Minute (WPM) and pause distribution. Penalty applied if $ ext{WPM} < 90$ or $>180$. |
| **Composite Score** | Deterministic PHP | $ ext{Overall} = 0.35 imes ext{Accuracy} + 0.35 imes ext{Clarity} + 0.30 imes ext{Fluency}$. Scaled down if Accuracy $< 85\%$. |

## 3. Conversational AI Mock Interview Engine

The interview engine ( `GeminiInterviewService.php` ) orchestrates technical and HR mock panels:

```
// System Prompt Topology for Gemini Voice Interviewer
$systemInstruction = "You are a Senior Principal Engineer conducting an interview for the role of {$role}.
Analyze the student's spoken input. If their answer is vague or misses algorithmic complexity, probe deeper.
Evaluate across 4 distinct rubrics:
1. Technical Correctness (35%)
2. Problem Solving & Edge Cases (30%)
3. Communication Clarity (20%)
4. Code Quality & Modularity (15%)";
```

## 4. Launch Safeguard: Paid Gemini API Key Isolation

> **Production Quota Warning: Free Tier vs Paid Tier**
> Development keys on the Google AI Studio free tier are capped at **15 requests/minute** across the entire project. A single interview or speaking attempt executes 2 parallel requests. In production, `GEMINI_API_KEY` must be provisioned on a Pay-As-You-Go Google Cloud project to prevent HTTP 429 quota exhaustion during semester batches.

## 1. Proctoring Telemetry & Verification Stream

The anti-cheat proctoring engine operates as a continuous state machine monitoring active candidate sessions during timed examinations. Signals are captured by client-side event listeners and validated server-side.

| PROCTORING SENSOR | CLIENT CAPTURE MECHANISM | SERVER VALIDATION & ENFORCEMENT ACTION |
|---|---|---|
| **Tab Blur / Window Switch** | `window.onblur` & `document.visibilitychange` | Server logs timestamp. Increments strike counter; triggers Strike 1 or Strike 2 alert modal. |
| **Fullscreen Escape** | `document.fullscreenElement === null` | Immediate fullscreen warning. 10-second countdown to return or strike is permanently committed. |
| **Webcam Face Verification** | Periodic WebRTC canvas snapshot (every 30s) | AI face detection flags: *Zero faces* (candidate left seat) or *Multiple faces* (collusion). |
| **DevTools & Inspect Intercept** | Keyboard listener on `F12`, `Ctrl+Shift+I` | Prevents opening developer console; logs attempted inspection event to proctoring log. |

## 2. The 3-Strike Security State Machine

**Strike 1**
INITIAL WARNING
Yellow alert banner. Candidate alerted. Recorded in audit ledger.

**Strike 2**
FINAL WARNING
Orange blocking modal: "One additional violation will submit your test".

**Strike 3 (Lockout)**
SESSION TERMINATED
Automatic submission of on-screen code. Test session permanently locked.

## 3. Administrative Reinstatement Architecture

**The One-Click Reinstatement Workflow ( `POST /api/proctoring/reinstate` )**

When a candidate is locked out due to an accidental system glitch (e.g. OS popup), the College TPO can reinstate them from `/admin/proctoring`.

- TPO inspects the incident log and webcam snapshot.
- TPO clicks `Reinstate Student` and enters a mandatory justification reason.
- The server resets strikes to 2, clears the lock state, and logs the action in the immutable audit ledger. The candidate resumes testing without losing previous answers.

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## 1. Spaced Repetition (SRS): The 6-Box Leitner Model

Vocabulary retention is driven by `VocabularySrsService.php`, implementing a 6-box Leitner scheduling model. Progress is tracked on the `vocabulary_word_progress` table per student, per word.

| LEITNER BOX | REVIEW SCHEDULE | STATUS RENDERED | PROGRESSION & DEMOTION RULES |
|---|---|---|---|
| **Box 0** | Due Today | LEARNING | New word introduced or word demoted after wrong answer. |
| **Box 1** | Due in 1 Day | LEARNING | Correct answer on or after due date advances word from Box 0. |
| **Box 2** | Due in 3 Days | FAMILIAR | Advances upon correct answer when due. |
| **Box 3** | Due in 7 Days | FAMILIAR | Advances upon correct answer when due. |
| **Box 4** | Due in 16 Days | MASTERED | Qualifies toward "Words Mastered" metric on student profile. |
| **Box 5** | Due in 35 Days | MASTERED | Long-term memory retention state. |

> **The Demotion Penalty Rule**
> An incorrect answer drops a word **two boxes** (never below Box 0) and marks it due today. Practicing ahead of schedule counts toward streak and accuracy, but cannot rush a word up before its due date.

## 2. Soft Skills Library: 70 Pre-Built Examination Sets

AptRun ships with 70 pre-seeded, standardized examination papers (Sets 1 to 70), each comprising 25 questions (1,750 questions total). The schema is managed via `ManagesSoftSkillQuestions` concern:

| QUESTION CATEGORY | QUESTIONS / SET | SYLLABUS BLUEPRINT & ALGORITHMIC DISTRIBUTION |
|---|---|---|
| **Quantitative Aptitude** | 8 Questions | Arithmetic, Time & Work, Speed-Distance, Profit & Loss, Permutations, Probability, Geometry. |
| **Logical Reasoning** | 7 Questions | Syllogisms, Blood Relations, Coding-Decoding, Seating Arrangements, Pattern Series. |
| **Verbal Ability / English** | 6 Questions | Error Spotting, Sentence Improvement, Para Jumbles, Reading Comprehension, Contextual Vocab. |
| **Situational Judgement** | 4 Questions | Workplace ethics, Conflict resolution, Cross-functional communication, Priority triage. |

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## 1. Payment Gateway Architecture: PayU & Razorpay

AptRun implements a dual-gateway purchase architecture ( `CheckoutService.php` ). PayU Hosted Checkout serves as default ( `PAYMENT_GATEWAY=payu` ), with Razorpay available as an instant zero-code alternative.

| GATEWAY PARAMETER | PAYU IMPLEMENTATION (DEFAULT) | RAZORPAY IMPLEMENTATION (ALTERNATIVE) |
|---|---|---|
| **Checkout Workflow** | Hosted Checkout (Redirect via form post to PayU). | Modal Popup (Client-side Razorpay.js checkout modal). |
| **Return URL Callback** | `POST /api/payu/return` (max 50 chars hostname). | Handled client-side; verified via server-side signature check. |
| **Webhook Endpoints** | `POST /api/webhooks/payu?token=<TOKEN>` | `POST /api/webhooks/razorpay` (HMAC SHA256 verified). |
| **Reconciliation Cron** | `php artisan payments:reconcile` (Scheduled every 15 minutes; queries gateways for pending orders). | |

## 2. Real-Time Plan Signal Polling ( `frontend/src/lib/planSync.ts` )

To avoid pinning persistent WebSocket worker processes on PHP-FPM, open dashboards poll a lightweight signature endpoint:

```
// Signal Generation in PlanSignalService.php
$signature = hash('sha256', implode('|', [
    $user->id,
    $subscription?->plan_id,
    $subscription?->ends_at?->timestamp,
    $subscription?->plan?->updated_at?->timestamp
]));
return response()->json(['signal' => $signature]);
```

**Client-Side Event Firing on Signature Drift**

The client poller ( `planSync.ts` ) requests `GET /api/me/plan-signal` every 30 seconds and on window focus. If the returned signature differs from the local cache, it dispatches `SUBSCRIPTION_CHANGED_EVENT` . All sidebar locks, plan badges, and usage meters refetch immediately without reloading the page.

## Standard Operating Procedures (SOPs) for Making Changes

Follow these architectural patterns to modify or extend platform features without breaking existing tenant boundaries:

### SOP 1: How to Add a New Plan or Modify Entitlements

1
Database Migration & Model Casts
If adding a new limit column, create migration: `php artisan make:migration add_xyz_to_plans_table`. Add column to `Plan::$fillable` and `Plan::casts()`.
2
Register in `PlanSeeder.php` & User Model
Update `PlanSeeder.php` with default limit for Coder, Expert, and College rows. Update `User::effectiveEntitlements()` to expose the new key in the resolved array.
3
Enforce in Controller & Expose on Sidebar
Check limit in domain controller using `$user->effectiveEntitlements()['xyz']`. If locking a sidebar route, update `frontend/src/components/dashboard/studentNav.ts`.

### SOP 2: How to Add a New Engineering Discipline / Branch

1. Seed the discipline in `disciplines` table (e.g. Chemical Engineering: `code = 'chem'`).
2. Add canonical branch aliases in `BranchService.php` ("CHEM", "Chemical", "CHE").
3. Upload question banks via Mellow Ops importer ( `/admin/core-prep` ).
4. Publish at least one timed blueprint test.
5. Toggle the branch **Live** in Superadmin Master Control. This automatically stamps `live_at` and initiates the 30-day ARS scoring grace period.

### SOP 3: How to Swap AI Model Providers (Gemini → Claude / OpenAI)

All AI calls route through dedicated services: `GeminiInterviewService` and `GeminiSpeakingPassageService`. To introduce Anthropic Claude or OpenAI:
1. Create an `AiProviderInterface` with methods: `transcribeAudio()`, `scoreRubrics()`, `generatePassage()`.
2. Implement `ClaudeAiProvider.php` using Guzzle HTTP client.
3. Bind the implementation in `AppServiceProvider.php` based on `.env` variable `AI_PROVIDER=claude`.

## 1. Technical Debt & Recommended Refactoring Items

The following architectural enhancements are recommended for implementation to improve maintainability and horizontal scalability:

| SUBSYSTEM | CURRENT IMPLEMENTATION | RECOMMENDED REFACTORING / IMPROVEMENT |
|---|---|---|
| **Plan Entitlements** | Discrete boolean and integer columns on `plans` table. | Migrate to a single `features (JSONB)` column with JSON schema validation to allow adding new feature flags without database migrations. |
| **Workspace Volume Retention** | Docker volumes ( `wsvol-<n>` ) remain on disk indefinitely after subscription lapses. | Implement an automated cleanup cron ( `workspaces:purge-expired` ) to archive volumes to S3 and delete local volumes 90 days after plan lapse. |
| **Queue Processing** | Scheduler cron runs `queue:work --stop-when-empty` every minute. | Deploy a dedicated Supervisor daemon running persistent queue workers ( `php artisan queue:work --sleep=3 --tries=3` ) for zero-latency email and event processing. |
| **Piston Judge Cluster** | Single Piston host server configured in `.env`. | Deploy an HAProxy load balancer in front of multiple Piston worker nodes with health checks to scale beyond 2,500 simultaneous submissions. |
| **Company Recruiter RBAC** | Single corporate user login per company tenant. | Introduce `company_memberships` table with fine-grained permissions (e.g. *Hiring Lead*, *Interviewer*, *Reviewer*). |

## 2. Production Readiness Checklist for Senior Developers

**1. Cache & Route Optimization**
Ensure `php artisan config:cache`, `php artisan route:cache`, and `php artisan view:cache` are executed on every production deployment.

**2. Payment Webhook Token Verification**
Verify that `PAYU_WEBHOOK_TOKEN` and `RAZORPAY_WEBHOOK_SECRET` are set in production `.env` to reject forged payment confirmation webhooks.

**AptRun™ Enterprise Architecture · Product of Mellow Vault (Prayukti Development Pvt. Ltd. · CIN: U62099UP2024PTC207212)**
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